

Root-sensitive comparison beyond fixed-proportion degree windows

Split-Zero research continuation

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Abstract

An explicit Laguerre evaluation bound controls the whole inner integration region for every complex polynomial direction. Combined with the literal opposite-root factorization, it gives a finite full-form comparison with error $O(qR^2(r+q)^2/q^4)$ at relation degree r , with all constants and finite guards retained. A separate Jensen argument supplies an all-degree monic reference lower bound. The original arithmetic monic minima, adjacent quotient ratios, both contraction factors and changed four endpoints then give a lower bound on the canonical allowance on genuinely super-proportional bands. The original source and geometric programme are not replaced, and no new arithmetic zero packet, formal certificate or RH conclusion is asserted.

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1 A root-sensitive full-form comparison at growing polynomial degree

Scope and provenance. This is a new written continuation of the supplied 680-page edition, not a claim that the Riemann hypothesis has been proved or disproved. It addresses the growing-degree comparison requested in the current continuation prompt. The original nineteen files are preserved byte-for-byte in the accompanying archive. Its source map and historical verification reports remain inherited records. The new estimates below have complete proofs and separately labelled finite checks, but no new Lean certificate or independent second-reader acceptance is claimed. The fixed-proportion results of PR37 are retained; none is extrapolated by changing a bounded equilibrium parameter into a moving one.

The principal advance is a full-coefficient comparison between multiplication by the original root polynomial and multiplication by the monomial of the same degree. Its error is explicit in the relation degree. The comparison includes the entire inner integration region, carries into the original monic minima, and gives a further, genuinely growing range for the canonical allowance. It does not evaluate the separate mixed allocations or their common flag profile.

1.1 Original arithmetic input and a finite comparison theorem

Retain a fixed hypothetical complete-order quartet and its original source:

$$\begin{aligned}
h(z) &= \prod_{\rho \in \{1/2 \pm \delta \pm i\gamma\}} (z - \rho)^m, \quad 0 < \delta < 1/2, \quad \gamma > 2, \quad m \in \mathbb{N}_{>0}, \\
k &= 4l + 1 \geq 5, \quad e_k = 1 + k(m - 1), \quad q = e_k(k + 1)^2, \quad c = k/2, \\
\chi(S) &= \prod_{a,b=0}^k (S - c - (2a - k)\delta - i(2b - k)\gamma)^{e_k}, \\
R = R_k &= k\sqrt{\delta^2 + \gamma^2}, \quad d\mu_k(y) = w_h^{*k}(y) dy, \quad w_h(y) = |(2\xi/h)(1/2 + iy)|^2/(2\pi).
\end{aligned} \tag{RGC1}$$

The monic polynomial in the real coordinate is $Q(y) = i^{-q}\chi(c + iy)$. Its q roots, with multiplicities, occur in opposite pairs and have modulus at most R . Since q is a multiple of four here, $i^{-q} = 1$, but the coordinate isometry is retained rather than assumed.

The fixed Gamma reference and density constants are exactly those of AMT:

$$\begin{aligned}
d\sigma(y) &= s(y) dy = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad \alpha = \pi/2, \\
c_- e^{-\alpha|y|}(1 + |y|)^{-1/2} &\leq s(y) \leq c_+ e^{-\alpha|y|}(1 + |y|)^{-1/2}, \\
c_- &= \sqrt{2}e^{-7/6}, \quad c_+ = \sqrt{2\sqrt{5}}e^{2/3}, \quad g_n = \sqrt{2\pi} \frac{(2n)!}{4^n}.
\end{aligned} \tag{RGC2}$$

The formulas below first apply to any even degree $q \geq 2$, any monic opposite-paired Q with radius R , and every integer $r \geq 0$. They are not restricted to $r \leq q$.

For $\deg P \leq r$ set

$$H_0(P) = \int_{\mathbb{R}} |y|^{2q} |P(y)|^2 d\sigma(y), \quad H_Q(P) = \int_{\mathbb{R}} |Q(y)P(y)|^2 d\sigma(y). \tag{RGC3}$$

Introduce the explicit finite quantities

$$\begin{aligned}
s_q &= 2q + 1, \quad D_{q,r} = \frac{4r + 4q + 2}{\alpha}, \quad Y_{q,r} = \frac{s_q^2}{32\alpha(r + s_q)}, \\
\eta_{q,r} &= \frac{c_+}{c_-} (1 + D_{q,r}) s_q (3/8)^{s_q}, \quad \rho_{q,r} = R/Y_{q,r}.
\end{aligned} \tag{RGC4}$$

Here s_q is not a source order, $D_{q,r}$ is a scalar moment bound, and $Y_{q,r}$ is an integration cutoff, not a new arithmetic coordinate.

Theorem RGC.1 (Finite full-form comparison). *Assume*

$$2R \leq Y_{q,r}, \quad \eta_{q,r} < 1. \tag{RGC5}$$

Then for every complex polynomial P of degree at most r ,

$$\begin{aligned}
(1 - \rho_{q,r}^2)^q (1 - \eta_{q,r}) H_0(P) &\leq H_Q(P), \\
H_Q(P) &\leq ((1 + \rho_{q,r}^2)^q + \eta_{q,r}) H_0(P).
\end{aligned} \tag{RGC6}$$

In particular, with

$$E_{q,r}^{\text{root}} = -q \log(1 - \rho_{q,r}^2) - \log(1 - \eta_{q,r}), \tag{RGC7}$$

one has $e^{-E_{q,r}^{\text{root}}} H_0 \leq H_Q \leq e^{E_{q,r}^{\text{root}}} H_0$. If $\eta_{q,r} \leq 1/2$, then

$$E_{q,r}^{\text{root}} \leq \frac{4}{3} q \rho_{q,r}^2 + 2\eta_{q,r} = \frac{4096\alpha^2 q R^2 (r + s_q)^2}{3s_q^4} + 2\eta_{q,r}. \tag{RGC8}$$

1.2 The complete inner-region bound

The point requiring a proof is control of every polynomial direction near the roots, not just a monic trial or an equilibrium support. Consider on the positive half-line the auxiliary integral

$$N_q(p) = \int_0^\infty |p(t)|^2 t^{2q} e^{-\alpha t} dt.$$

This is an integration comparator for each of $P(t)$ and $P(-t)$, not a replacement quotient or source metric.

Lemma RGC.2 (Laguerre moment and evaluation bounds). *For every $\deg p \leq r$,*

$$\int_0^\infty t |p(t)|^2 t^{2q} e^{-\alpha t} dt \leq D_{q,r} N_q(p), \quad (\text{RGC9})$$

and for every $t \geq 0$,

$$|p(t)|^2 \leq K_{q,r}(0,0) e^{2\alpha r t/s_q} N_q(p), \quad K_{q,r}(0,0) = \frac{\alpha^{s_q}}{(s_q - 1)!} \binom{r + s_q}{r}. \quad (\text{RGC10})$$

Proof. Put $\beta = 2q$. Rodrigues' formula and l integrations by parts give the Laguerre norm

$$\int_0^\infty [L_l^{(\beta)}(\alpha t)]^2 t^\beta e^{-\alpha t} dt = \alpha^{-\beta-1} \frac{\Gamma(l + \beta + 1)}{l!}.$$

For clarity, orthogonality follows because all derivatives of a polynomial of degree below l vanish after l integrations. The boundary terms vanish at zero since β is a nonnegative integer, and at infinity by the exponential. The leading coefficient is $(-1)^l/l!$ in the scaled variable; using it in the same integration gives the displayed norm. The three-term recurrence follows also by coefficient comparison in

$$L_l^{(\beta)}(x) = \sum_{j=0}^l (-1)^j \binom{l + \beta}{l - j} \frac{x^j}{j!}.$$

In the normalized basis, multiplication by t has diagonal $(2l + \beta + 1)/\alpha$ and adjacent entries with absolute value $\sqrt{(l + 1)(l + \beta + 1)}/\alpha$. The downward entry is at most $(l + \beta/2)/\alpha$ and the upward entry at most $(l + 1 + \beta/2)/\alpha$. The absolute row sum of its compression through degree r is thus at most $(4r + 2\beta + 2)/\alpha = D_{q,r}$. Its quadratic form is the integral in (RGC9); no next-degree output has been discarded in a claimed norm bound. We are bounding a first moment, so this compression is exactly the required quadratic form.

The finite series also gives

$$\frac{|L_l^{(\beta)}(\alpha t)|}{L_l^{(\beta)}(0)} \leq \sum_{j=0}^l \frac{l(l-1)\cdots(l-j+1)}{(\beta+1)_j} \frac{(\alpha t)^j}{j!} \leq e^{l\alpha t/(\beta+1)}.$$

Cauchy–Schwarz in the complete degree- r normalized basis proves the pointwise bound. Its value at zero is

$$\frac{\alpha^{\beta+1}}{\beta!} \sum_{l=0}^r \binom{l + \beta}{l} = \frac{\alpha^{\beta+1}}{\beta!} \binom{r + \beta + 1}{r},$$

using the finite binomial summation identity. This proves (RGC10). These classical Laguerre formulas agree with DLMF §§18.3, 18.5.12, 18.6.1 and 18.9. $\square$

By (RGC2), $s(t) \geq c_- e^{-\alpha t}/(1 + t)$ for $t \geq 0$. Cauchy–Schwarz and (RGC9) therefore give

$$\int_0^\infty t^{2q} |p(t)|^2 s(t) dt \geq \frac{c_-}{1 + D_{q,r}} N_q(p). \quad (\text{RGC11})$$

Indeed $N_q(p)^2$ is at most the product of the same integrals with the factors $(1+t)^{-1}$ and $(1+t)$ inserted. Apply this separately to $p(t) = P(t)$ and $p(t) = P(-t)$ and add. This retains both half-lines and does not require real coefficients or a parity decomposition of a mixed matrix.

If $P = 0$, both forms and both inner integrals vanish. In the quotient estimate below take $P \neq 0$; after multiplication by $H_0(P)$, its undivided form holds also for $P = 0$.

For any $Y > 0$, every root of Q has modulus at most R , so $|Q(y)| \leq (Y + R)^q$ on $|y| \leq Y$. Combining (RGC10)–(RGC11) and $s(y) \leq c_+$ yields

$$\frac{\int_{|y| \leq Y} |Q(y)P(y)|^2 d\sigma(y)}{H_0(P)} \leq \frac{c_+}{c_-} (1 + D_{q,r}) Y K_{q,r}(0,0) (Y + R)^{2q} e^{2\alpha r Y / s_q}. \quad (\text{RGC12})$$

There is no extra factor of two: the two half-line numerator estimates are summed along with their two lower norm estimates. The same formula with $R = 0$ controls the inner part of $H_0(P)$.

Take $Y = Y_{q,r}$ and $R \leq Y/2$. Since

$$\binom{r + s_q}{r} \leq \frac{(r + s_q)^{s_q}}{s_q!}, \quad s_q! \geq (s_q/e)^{s_q},$$

one has

$$Y K_{q,r}(0,0) (Y + R)^{2q} e^{2\alpha r Y / s_q} \leq s_q \left(\frac{3e^{33/16}}{64} \right)^{s_q} < s_q (3/8)^{s_q}.$$

The factorial bound follows from the integral lower bound for $\sum \log j$. For the last strict inequality, the positive series $\log 2 = 2 \sum_{j \geq 0} (1/3)^{2j+1} / (2j+1)$ gives $\log 2 > 56/81$, so $\log 8 > 56/27 > 33/16$. Thus both inner integrals are at most $\eta_{q,r} H_0(P)$. This proves the full inner estimate for arbitrary r , rather than assuming that P avoids the root region.

1.3 Outer region, finite proof and degree range

Pair the opposite roots as $\pm z_a$. For $|y| > Y > R$,

$$\frac{|Q(y)|^2}{|y|^{2q}} = \prod_{a=1}^{q/2} |1 - z_a^2/y^2|^2.$$

Therefore

$$(1 - R^2/Y^2)^q \leq \frac{|Q(y)|^2}{|y|^{2q}} \leq (1 + R^2/Y^2)^q. \quad (\text{RGC13})$$

Integrating the lower inequality on the outer region and subtracting the bounded inner part of H_0 proves the first line of (RGC6). Adding the bounded inner part of H_Q to the outer upper inequality proves its second line. This proves Theorem RGC.1. For its symmetric version use

$$(1 + u)^q + \eta \leq \frac{(1 - u)^{-q}}{1 - \eta} \quad (0 \leq u < 1, \ 0 \leq \eta < 1).$$

Finally $-\log(1 - u) \leq u/(1 - u)$, with $u \leq 1/4$, and $-\log(1 - \eta) \leq 2\eta$ for $\eta \leq 1/2$, prove (RGC8).

The explicit radius guard is

$$64\alpha R(r + 2q + 1) \leq (2q + 1)^2. \quad (\text{RGC14})$$

This undivided guard includes $R = 0$ and is equivalent to $2R \leq Y_{q,r}$ for every $R \geq 0$. For the original arithmetic radius $R_k > 0$ it also gives the displayed cutoff after division by $64\alpha R_k$. For fixed arithmetic data, $R = O_h(k)$ and $q \geq (k + 1)^2$. Thus this admits genuinely growing ratios r/q . More precisely, for any sequence of cutoffs $r = r_k$ satisfying

$$\frac{R_k(r_k + q)}{q^2} \longrightarrow 0, \quad (\text{RGC15})$$

the guards eventually hold and $E_{q,r_k}^{\text{root}} = o(q)$. Here $r_k = O_h(q^2)$ eventually, so the prefactor in η_{q,r_k} is polynomial and its exponential term dominates it. This assertion concerns a full-form comparison, not a uniform extension of the equilibrium profile ψ .

1.4 A non-equilibrium reference lower bound at every relation degree

We next calculate a lower reference bound without a compact equilibrium parameter rectangle. Define

$$\nu_r^0 = \inf_{P \text{ monic, deg } P=r} H_0(P), \quad T_r^0 = \nu_r^0 / g_{q+r}, \quad n = q + r.$$

Theorem RGC.3 (Uniform monic reference lower bound). *For all integers $q \geq 1$ and $r \geq 0$,*

$$\log T_r^0 \geq F_{q,r} + \log C_0, \quad F_{q,r} = \frac{q^2}{4(q+r) \log(8(q+r)/q)}, \quad C_0 = \frac{7c_-}{32\sqrt{6}} e^{-1/24}. \quad (\text{RGC16})$$

The lower bound is not claimed to be a sharp reference asymptotic.

Proof. Put $x = q/n$, $L = \log(8/x)$, and $a = x/(4L)$; then $0 < a < 1/8$. For $0 < a \leq 1/8$, set

$$\bar{E}(a) = 1 + a^2 + \frac{1}{2}a^2 \log(2/a).$$

The elementary elliptic integral

$$E(a) = \int_0^{\pi/2} \sqrt{\sin^2 t + a^2 \cos^2 t} dt$$

satisfies $E(a) \leq \bar{E}(a)$. To see this, split at $t_0 = \arcsin a$. On the first interval the excess over $\sin t$ is at most $a \cos t$, with integral a^2 . On the second interval rationalization bounds that excess by $a^2 \cos^2 t / (2 \sin t)$; its integral is

$$\frac{a^2}{2} \left[\log \frac{1 + \sqrt{1 - a^2}}{a} - \sqrt{1 - a^2} \right] \leq \frac{a^2}{2} \log(2/a).$$

Also $\bar{E}(a) < 2$. Indeed $a^2 \log(2/a)$ increases on $(0, 1/8]$ and $\log 16 < 3$, giving $\bar{E} \leq 1 + 1/64 + 3/128 < 2$.

Let $b = 2n/\bar{E}(a)$ and use the probability measure

$$d\nu(y) = \frac{|y| dy}{\pi \sqrt{(y^2 - a^2 b^2)(b^2 - y^2)}} \quad \text{on } [-b, -ab] \cup [ab, b].$$

Its square-pushforward is the arcsine probability on $[a^2 b^2, b^2]$. The following three formulas follow by that substitution and circle Jensen:

$$\begin{aligned} \int \log |P| d\nu &\geq r \log \frac{b\sqrt{1 - a^2}}{2}, \\ \int \log |y| d\nu &= \log \frac{b(1 + a)}{2}, \\ \int \log(d\nu/dy) d\nu &= \log \frac{2}{\pi b(1 - a)}. \end{aligned}$$

For completeness, symmetry makes the integral contributed by a root z of P equal to one half of the arcsine potential at z^2 . Write $u = C + D \cos \theta$ for the square interval. Factoring the quadratic in $e^{i\theta}$, whose roots have product one, shows that its circle-logarithmic mean is at least $\log(D/2)$; this gives the first inequality root by root. The mean of $\log u$ gives the second formula. The two endpoint logarithmic integrals are each $\log((b^2 - a^2 b^2)/4)$; substitution into the density gives the third. Logarithmic singularities on the bands are integrable. Furthermore

$$\alpha \int |y| d\nu = bE(a) \leq 2n.$$

Probability Jensen applied to the norm integrand divided by $d\nu/dy$, together with (RGC2), now gives for every monic P

$$H_0(P) \geq \frac{c_- \pi b(1-a)}{2\sqrt{1+b}} n^{2n} e^{-2n} \exp\left(2q \log(1+a) + r \log(1-a^2) - 2n \log \bar{E}(a)\right). \quad (\text{RGC17})$$

No reference minimizer has been guessed.

For the exponent use $\log(1+a) \geq a - a^2/2$, $\log(1-a^2) \geq -4a^2/3$, and $\log \bar{E}(a) \leq a^2 + \frac{1}{2}a^2 \log(2/a)$. It is at least

$$2qa - na^2(10/3 + \log(2/a)).$$

Since $\log(2/a) = L + \log L$ and $L > 2$, one has $10/3 + \log(2/a) \leq 2L + 7/3 \leq 4L$. Substituting $a = q/(4nL)$ gives the lower exponent $q^2/(4nL) = F_{q,r}$.

The positive-real Stirling remainder gives

$$g_n \leq 2\pi\sqrt{2n} n^{2n} e^{-2n} e^{1/(24n)}.$$

Here $n \leq b \leq 2n$, $1-a \geq 7/8$, and $\sqrt{2n(1+b)} \leq \sqrt{6}n$. Dividing (RGC17) by this upper bound for g_n leaves the factor at least $7c_- e^{-1/24}/(32\sqrt{6}) = C_0$. Take the same monic infimum to prove (RGC16). $\square$

1.5 Transport through the original arithmetic monic and quotient maps

The original AMT full-form bounds, valid at each polynomial degree n , are

$$\frac{a_k}{\mathfrak{D}_n} \|F\|_\sigma^2 \leq \|F\|_{\mu_k}^2 \leq A_k \|F\|_\sigma^2, \quad \mathfrak{D}_n = [1 + 4(n + 21 + 4m)^2]^{21+4m}. \quad (\text{RGC18})$$

The constants remain

$$\begin{aligned} B_h &= 42 + 8m, \quad p_h = 8m - 9/2, \quad \vartheta_h = \int_{-1}^1 w_h(y) dy, \quad M_\alpha = \int e^{\alpha y} w_h(y) dy, \\ a_k &= \frac{c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} (k-2)^{-B_h}}{c_+ 2^{B_h/2}}, \quad A_k = \frac{C_h^{\text{tilt}}}{c_-} k^{p_h+1} M_\alpha^{k-1}. \end{aligned} \quad (\text{RGC19})$$

The original positive constants c_h, C_h^{tilt} are the supplied arithmetic density-envelope constants. They are inputs from AMT/TW, not newly evaluated numerical parameters. In particular, for fixed h ,

$$\Lambda_{k,n} := \log(A_k \mathfrak{D}_n / a_k) = O_h(k + \log(n+1)).$$

Let ω_n be the original arithmetic monic norm and $\nu_j^\chi = \inf_{P \text{ monic, deg } P=j} \|\chi P\|_{\mu_k}^2$. In this formula the variable is the original S ; the map $P(S) \mapsto i^{-j} P(c + iy)$ is the degree- j monic coefficient isometry. It changes no quotient or arithmetic unit. The original exact sequence is

$$0 \longrightarrow \mathcal{P}_j \xrightarrow{\times \chi} \mathcal{P}_{q+j} \xrightarrow{J_{q+j}} E = \mathbb{C}[S]/(\chi) \longrightarrow 0. \quad (\text{RGC20})$$

The source realization and its full observation remain $\mathcal{V}(P) = P(\sum D_i) F_h^{\otimes k}$ and $J^{(k)} \mathcal{V} = \eta_{h,k} \pi_\chi$, with the entire Taylor unit inside $\eta_{h,k}$. Multiples of χ have their original theta primitives. The norm of a relation in (RGC20) is measured before its quotient value becomes the represented zero at the receiving support.

For $n = q+r$ under (RGC5), taking the same monic minima in Theorem RGC.1 and (RGC18) gives, for $0 \leq j \leq r$,

$$\left| \log T_j^{\text{ar}} - \log T_j^0 \right| \leq \Lambda_{k,n} + E_{q,r}^{\text{root}}, \quad T_j^{\text{ar}} = \nu_j^\chi / \omega_{q+j}. \quad (\text{RGC21})$$

The same comparison, with $\mathfrak{D}_d \leq \mathfrak{D}_n$, gives

$$\left| \log \frac{\beta_d}{d(d-1/2)} \right| \leq \Lambda_{k,n}, \quad \beta_d = \omega_d / \omega_{d-1}, \quad q \leq d \leq n. \quad (\text{RGC22})$$

Neither estimate is obtained by differentiating a determinant asymptotic.

The source/relation Gram determinant identity in (RGC20) proves, by taking one adjacent ratio,

$$d_{q+j} := \frac{V_{q+j}}{V_{q+j-1}} = \frac{\omega_{q+j}}{\nu_j^X} = (T_j^{\text{ar}})^{-1}. \quad (\text{RGC23})$$

At the original untilted source, the established total reflection identity then gives

$$\epsilon_{q+j-1}^2 = \beta_{q+j}(T_j^{\text{ar}} - 1) \begin{cases} 1, & j = 0, \\ 1 - (T_{j-1}^{\text{ar}})^{-1}, & j \geq 1. \end{cases} \quad (\text{RGC24})$$

This uses total parity only. It does not assert that the individual mixed kernel and boundary pairings vanish. The initial case uses J_{q-1} , which is an isomorphism, and no inverse at degree $q-2$.

Corollary RGC.4 (Finite growing-band allowance bound). *Set*

$$S_{k,q,r} = F_{q,r} + \log C_0 - \Lambda_{k,q+r} - E_{q,r}^{\text{root}}, \quad b_{k,q,r}^{\text{low}} = \frac{a_k}{A_k \mathfrak{D}_{q+r}} q(q-1/2). \quad (\text{RGC25})$$

If (RGC5) holds and $S_{k,q,r} > 0$, then

$$T_j^{\text{ar}} \geq e^{S_{k,q,r}} \quad (0 \leq j \leq r), \quad \min_{q-1 \leq N \leq q+r-1} \epsilon_N \geq \sqrt{b_{k,q,r}^{\text{low}}} e^{S_{k,q,r}/2} (1 - e^{-S_{k,q,r}}). \quad (\text{RGC26})$$

Proof. $F_{q,j}$ decreases with j . Equations (RGC16) and (RGC21) therefore give the first bound simultaneously. Equation (RGC22) gives the common lower bound for β_d . Insert both in (RGC24). The first row has one instead of two contraction factors and hence also obeys the displayed lower bound. $\square$

1.6 A genuinely super-proportional range

Corollary RGC.5. *At fixed complete arithmetic data, take $r_k = \lfloor qk^\theta \rfloor$. For a simple quartet, every fixed $0 < \theta < 2/3$ satisfies the guards and, eventually,*

$$\log \min_{q-1 \leq N \leq q+r_k-1} \epsilon_{h,k,N} \geq (1 - o_h(1)) \frac{q}{8(1 + k^\theta) \log(8(1 + k^\theta))}. \quad (\text{RGC27})$$

For every fixed multiplicity $m \geq 2$, the same statement holds for each fixed $0 < \theta < 4/3$. The right-hand side grows faster than $\log k$. Consequently the ratio of this minimum to the original polynomial exterior benchmark tends to infinity on these growing bands as well.

Proof. Write $q \asymp_h k^d$ with $d = 2$ for $m = 1$ and $d = 3$ for $m \geq 2$. Then $R = O_h(k)$, $\Lambda_{k,q+r_k} = O_h(k + \log q)$, and

$$\rho_{q,r_k} = O_h(k^{1+\theta-d}), \quad E_{q,r_k}^{\text{root}} = O_h(k^{2+2\theta-d}) + e^{-\Omega_h(q)}, \quad F_{q,r_k} \asymp_h \frac{k^{d-\theta}}{\log k}.$$

The stated ranges make $\theta < d-1$ and $3\theta < 2d-2$. Thus the radius guard holds, the tail is exponentially small, and both Λ and E^{root} are $o(F)$. It follows that $S_{k,q,r_k} = (1 - o_h(1))F_{q,r_k}$. The logarithm of b^{low} is $O_h(k + \log q) = o(F)$, and $\log(1 - e^{-S}) = o(1)$. Apply (RGC26). The floor changes F by a relative $o(1)$ only. Finally the exterior benchmark is $L_{h,k} = \delta q(k+1)/2$ on this subsequence, so $\log L_{h,k} = O_h(\log k)$. $\square$

This is a lower bound, not the sharp profile $q\psi(r/q)$ and not an extension of its fixed-rectangle error. In particular no claim is made at the endpoints $\theta = 2/3, 4/3$, for faster growing degree, or at a chosen exponential degree. The exact comparison theorem (RGC6) can be used over its full finite guarded range, even when the stronger dominance assumptions of Corollary RGC.5 do not hold.

1.7 Derive the changed four endpoints and complete action

For an arbitrary integer $r \geq 1$, give the rows $N = q - 1, \dots, q + r - 1$ weight one at the two ends and weight two in the interior. Their weights sum to $2r$. Define

$$\mathcal{J}_{q,r} = \sum_{N=q-1}^{q+r-1} w_N \log \epsilon_N^2, \quad \mathcal{B}_{q,r} = \log \frac{V_{q-1} V_q}{V_{q+r-1} V_{q+r}}, \quad \mathcal{W}_{q,r} = \log \frac{\omega_{q+r-1} \omega_{q+r}}{\omega_{q-1} \omega_q}. \quad (\text{RGC28})$$

Put $a_d = -\log(1 - d_d)$ and retain the precise penalty

$$\mathcal{P}_{q,r} = \sum_{j=0}^r w_{q+j-1} a_{q+j} + \sum_{j=1}^r w_{q+j-1} a_{q+j-1}. \quad (\text{RGC29})$$

The total coefficient is $4r - 1$. For $r \geq 2$ the coefficients are $3, 4, \dots, 4, 3, 1$ on $a_q, \dots, a_{q+r}$; for $r = 1$ they are $2, 1$. Taking logarithms in (RGC24) and telescoping the exact norm and volume ratios proves

$$\boxed{\mathcal{J}_{q,r} = \mathcal{B}_{q,r} + \mathcal{W}_{q,r} - \mathcal{P}_{q,r}}. \quad (\text{RGC30})$$

Under the guards of Corollary RGC.4,

$$0 \leq \mathcal{P}_{q,r} \leq -(4r - 1) \log(1 - e^{-S_{k,q,r}}). \quad (\text{RGC31})$$

The original two-window identity is recovered at $r = q$. The new four endpoints have been derived, not inserted into an old formula by analogy. The exact weighted action does not equate its geometric mean with its minimum.

For the monic part one also retains the finite AMT enclosure

$$\begin{aligned} -\Lambda_{k,q+r-1} - \Lambda_{k,q+r} &\leq \mathcal{W}_{q,r} - \mathcal{W}_{q,r}^\sigma \\ &\leq \Lambda_{k,q-1} + \Lambda_{k,q}, \end{aligned} \quad (\text{RGC32})$$

where $\mathcal{W}^\sigma$ is the identical four-norm expression with g_n . This comparison is taken after telescoping; it incurs four endpoint errors, not one source error for every row.

1.8 Support, interpretation, and remaining work

Every form inequality above holds before taking an observation or an affine minimum. It therefore passes to a specified restriction and to the minimum on a specified original fibre, with the inequality direction unchanged. The comparison of χP with $y^q P$ is a comparison of coefficient norms on the same domain $\mathcal{P}_r$. It is not an isomorphism between the arithmetic quotient and a homogeneous quotient. In particular the original square $G(\mathbb{Z}) \rightarrow G(\mathbb{C})$ over $\mathbb{Z} \rightarrow \mathbb{C}$, its infinite amplitude quotient, mixed support transports, full Taylor unit and all primary multiplicities remain upstream.

The original source map sends the relations $\chi \mathcal{P}_r$ to their actual theta boundaries. A boundary is sent to its receiving fibre's represented zero; external absence τ is not identified with that zero. Measuring the relation before this map is essential to (RGC23). No term of a mixed Gram is declared zero because its class becomes a boundary.

The growing-range theorem removes one concrete limitation of the earlier comparison: it controls both inner and outer regions at a specified moving polynomial cutoff. The further allowance lower bound uses only this new comparison and a uniform Jensen comparator. It shows that these additional canonical bands still do not yield a small allowance relative to the original exterior benchmark. This conclusion concerns the stated construction and regime. It neither settles the signed flag allocation nor rules out a quantitatively different same-class comparison in the broader tau programme.

No new claim about the current intrinsic allocation coefficient is made. Its surviving eight-matrix signed profile and all CPQ complementary returns remain those in the inherited continuation prompt. Improving them must be assessed through their actual receiving map; the cancellation in the canonical complete action is preserved in (RGC30).

References and exact inputs. The supplied `13_ARITHMETIC_MIXED_TRANSFER.tex`, AMT1–6, provides (RGC2), (RGC18) and (RGC19) at their stated original-source scope. The current cumulative file’s PCRP and RXUCT sections retain the older bounded-degree comparison and exact original ratio and radius identities. The new coefficient-degree extension and Jensen bound are proved in this section; their novelty relative to all literature is not asserted. Classical normalizations were checked against the following primary reference sections:

- National Institute of Standards and Technology. (n.d.). *Definitions; explicit representations; special values; recurrence relations*, DLMF §§18.3, 18.5.12, 18.6.1 and 18.9. <https://dlmf.nist.gov/18.5.E12>.
- National Institute of Standards and Technology. (n.d.). *Asymptotic expansions*, DLMF §5.11(i)–(ii), including the positive-real remainder for Stirling’s formula. <https://dlmf.nist.gov/5.11>.